ECE 303 P
Final Project: Frequency Hopping

Guided Study Questions (Due April 22nd)

1. What is BER? Is it the same as symbol error rate? If not, how are the two related? Why is BER plotted against E_b/N_0 rather than SNR, and what does E_b/N_0 represent physically?
2. How do you measure BER on a live FSK hardware link? Do you need a known reference sequence for this? What kind of sequence is used for this, and why is a pseudo-random sequence preferred over a fixed pattern?
3. What is FHSS and what physically happens to the transmitted signal hop by hop? It solves more than one problem; identify at least three problems that it solves. Give a real deployed system for each where that problem was the primary motivation for choosing FHSS.
4. What is the bandwidth of an FHSS signal? Define the two different bandwidths defined for FHSS, write the expression for each, and explain which one determines processing gain. How does processing gain affect a jammer's ability to disrupt the link?
5. What is the difference between fast and slow hopping? Define each precisely in terms of hop rate and bit rate. For each mode, what happens to a bit that lands on a jammed channel — and which mode recovers more gracefully from partial-band jamming and why?
6. Does hopping rate affect the signal bandwidth? Answer separately for instantaneous bandwidth and total hopping bandwidth. What happens spectrally at each hop transition, and does it get worse at higher hop rates? What hardware constraint sets a practical upper limit on hop rate?

Implementation: (Due April 29th)

1. Design a receiver that can receive FHSS FSK signal. Following abilities should be present in your code.
    a. Determine the frequency range over which the received signal seems to be hopping.
    b. Determine the hopping rate (classify whether it is slow or fast hopping).
    c. Given the hopping sequence and bit duration, demodulate the signal.
    d. Determine the received text.
    e. Identify any errors that may have occurred in the text.